

Introduction to Parallel Databases

1 Evolution and Motivation

Parallel machines are becoming quite common and affordable due to sharp drops in the prices of microprocessors, memory, and disks.

Trends in Computing

- **Hardware Proliferation:** Recent desktop computers increasingly feature multiple processors, a trend that is projected to accelerate.
- **Data Explosion:** Databases are growing larger as massive volumes of transaction data are collected and stored for analysis (Data Warehousing).
- **Multimedia Content:** Images and complex multimedia objects are increasingly integrated into modern databases.

2 Usage of Parallel Databases

Large-scale parallel database systems are increasingly utilized for critical enterprise tasks:

- **Data Volume Management:** Storing and managing petabytes of information effectively.
- **Decision-Support:** Processing complex, time-consuming queries that would be impractical on serial systems.
- **High Throughput:** Providing the necessary performance for intensive transaction processing systems.

3 Parallelism in Databases

Databases naturally lend themselves to parallelism due to their structured nature and high-level query languages.

Key Parallelism Mechanisms

- **I/O Parallelism:** Data can be partitioned across multiple disks to allow simultaneous read/write operations.
- **Relational Operation Parallelism:** Individual operations such as **Sort, Join, and Aggregation** can be executed in parallel.
- **Independent Processing:** Data can be partitioned so that each processor works independently on its own subset of data.

3.1 Why Parallelization is Easier in DBs

Queries are expressed in high-level languages like **SQL**, which are translated into relational algebra. This abstraction makes logical parallelization much easier compared to general-purpose code.

3.2 Query-Level Parallelism

Different queries can be run in parallel with each other. Concurrency control mechanisms (like locking and timestamping) ensure that conflicts between these parallel queries are handled correctly.

The shift towards parallel database architectures is driven by the necessity to process ever-growing datasets with high performance and reliability.